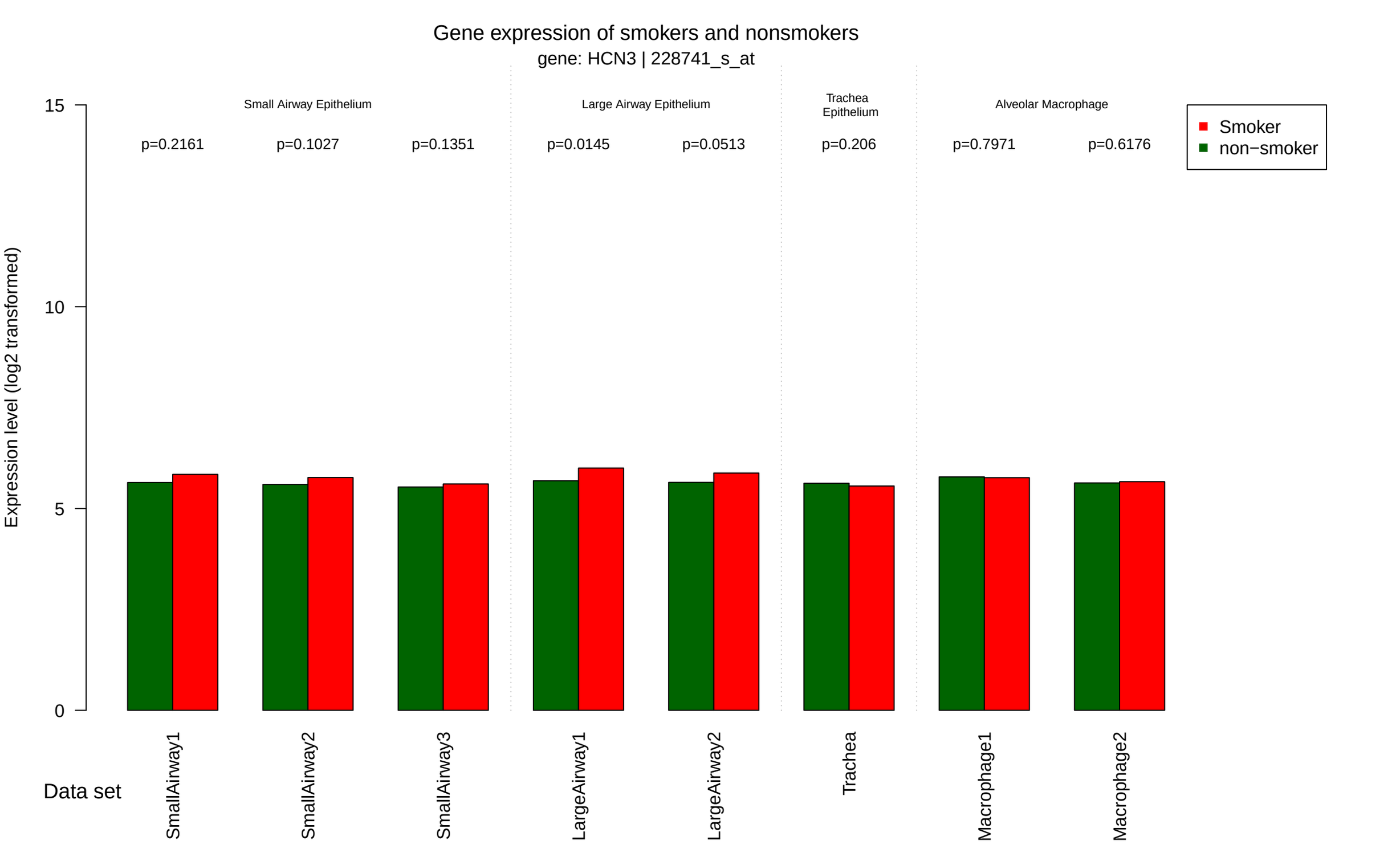
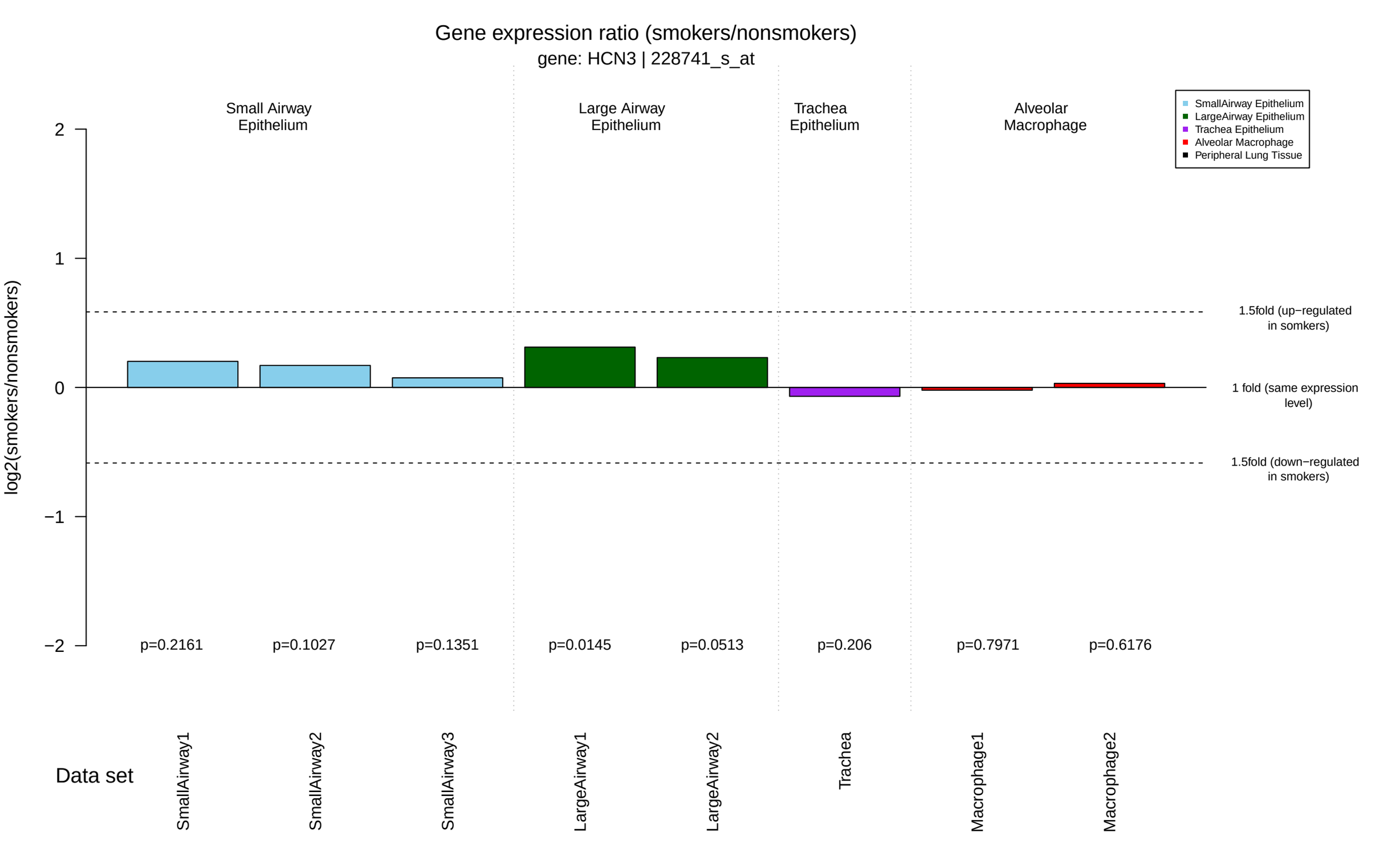
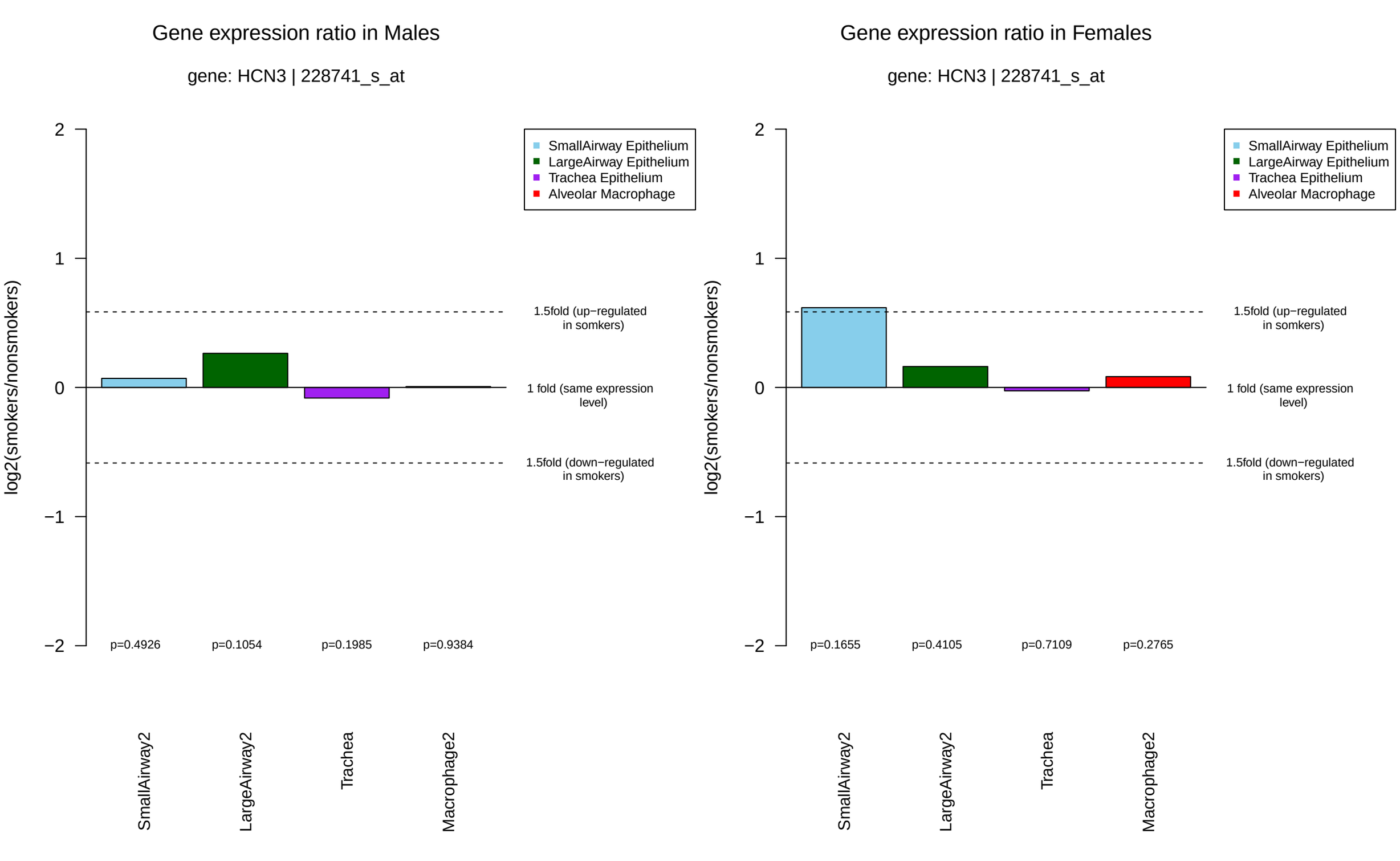
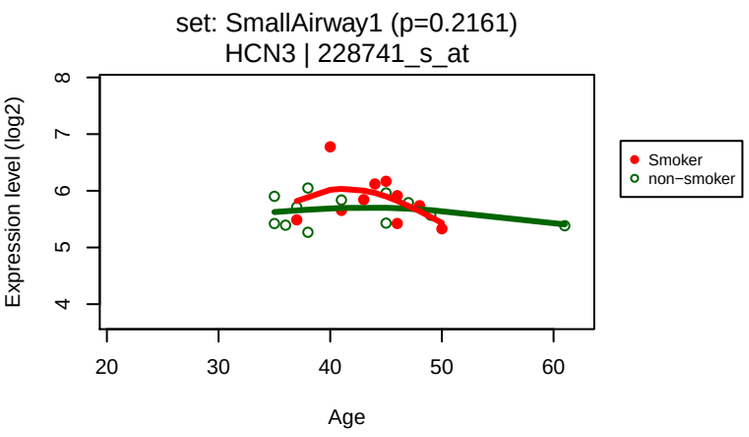
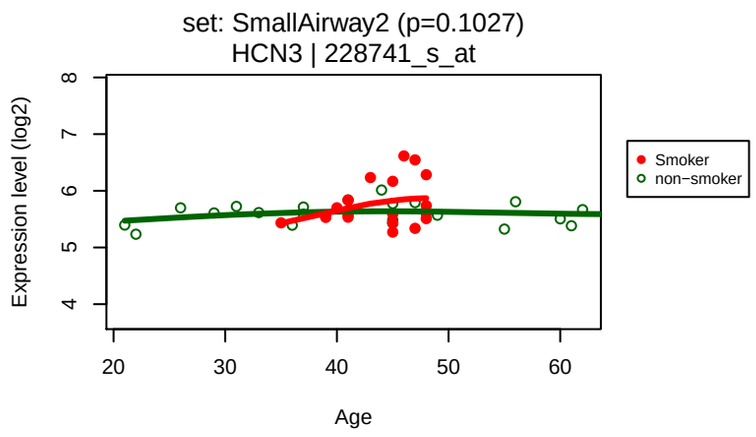
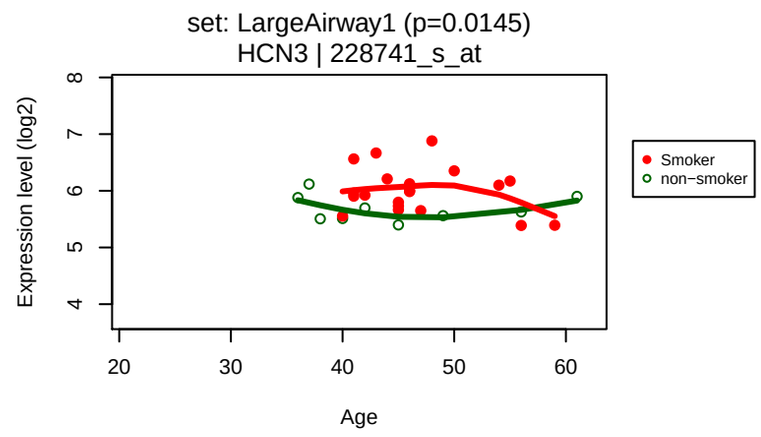

Gene expression of smokers and nonsmokers
gene: HCN3 | 228741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2161
p=0.1027
p=0.1351
p=0.0145
p=0.0513
p=0.206
p=0.7971
p=0.6176
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

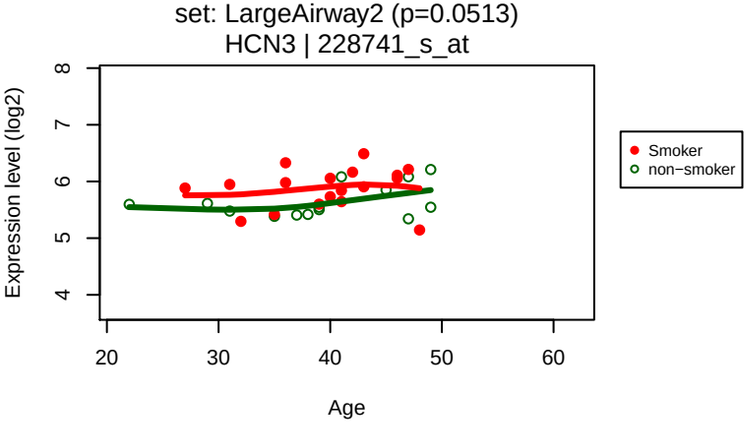

Gene expression ratio (smokers/nonsmokers)
gene: HCN3 | 228741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2161
p=0.1027
p=0.1351
p=0.0145
p=0.0513
p=0.206
p=0.7971
p=0.6176
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

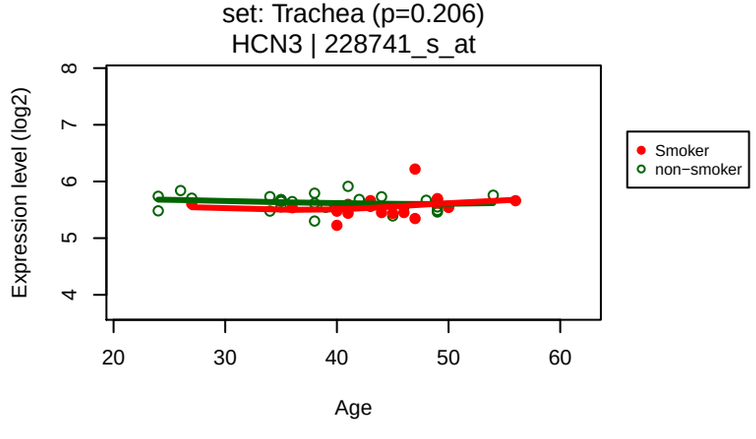

Gene expression ratio in Males
gene: HCN3 | 228741_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4926
p=0.1054
p=0.1985
p=0.9384
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HCN3 | 228741_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1655
p=0.4105
p=0.7109
p=0.2765
SmallAirway2
LargeAirway2
Trachea
Macrophage2

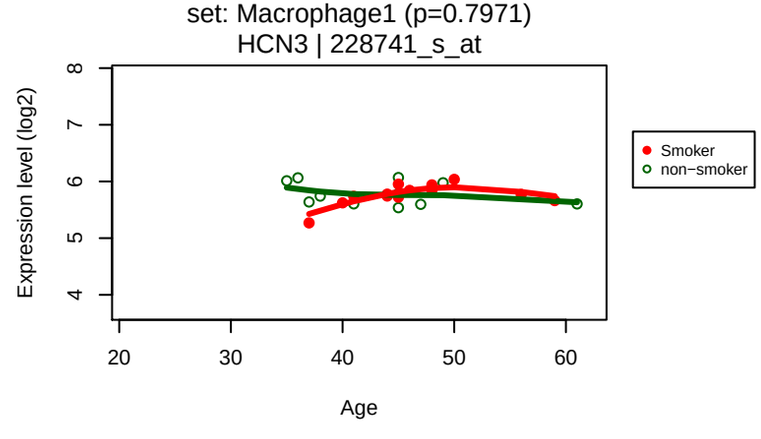

set: SmallAirway1 (p=0.2161)
HCN3 | 228741_s_at
Expression level (log2)
Age
Smoker
non−smoker

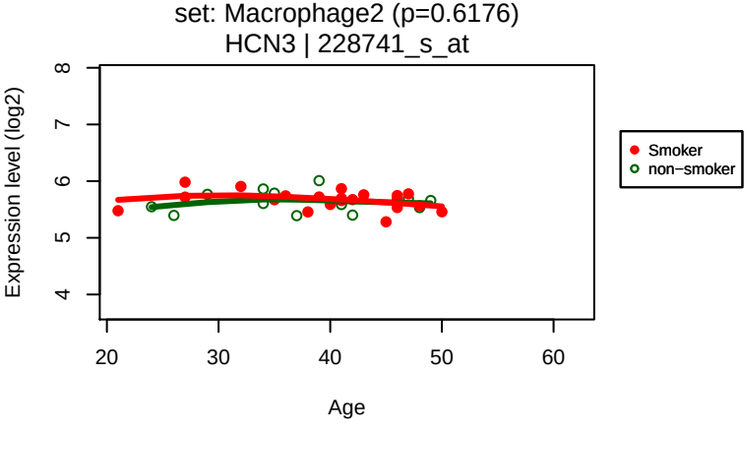

set: SmallAirway2 (p=0.1027)
HCN3 | 228741_s_at
Expression level (log2)
Age
Smoker
non−smoker

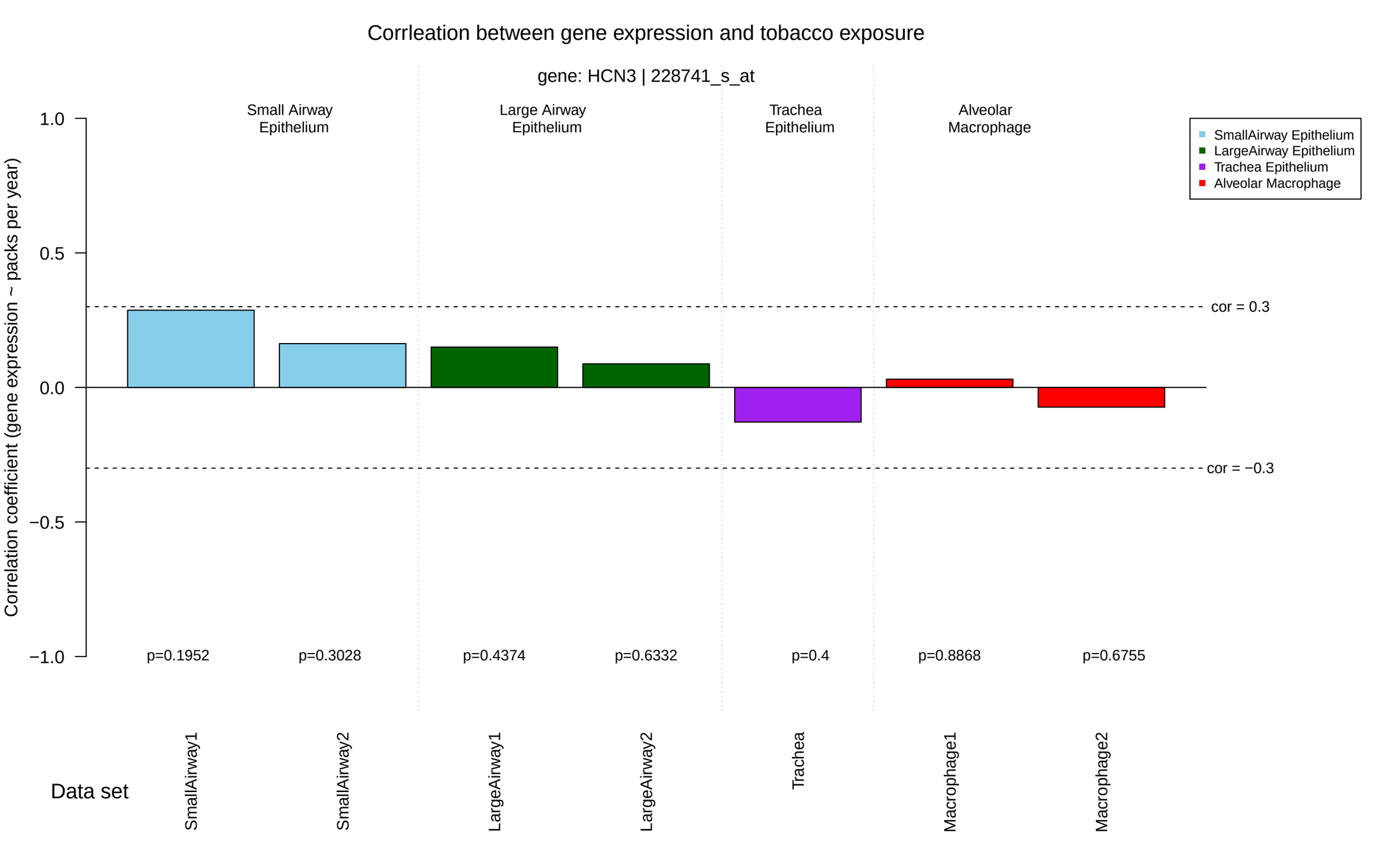

set: LargeAirway1 (p=0.0145)
HCN3 | 228741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0513)
HCN3 | 228741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.206)
HCN3 | 228741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7971)
HCN3 | 228741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6176)
HCN3 | 228741_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HCN3 | 228741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1952
p=0.3028
p=0.4374
p=0.6332
p=0.4
p=0.8868
p=0.6755
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2